Large language models as interactive cognitive agents: Cognitive surrender, augmentation, and psychopathology

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"*For this invention will produce forgetfulness in the minds of those who learn to use it, because they will not practice their memory. Their trust in writing, produced by external characters which are no part of themselves, will discourage the use of their own memory within them. You have invented an elixir not of memory, but of reminding; and you offer your pupils the appearance of wisdom, not true wisdom, for they will read many things without instruction and will therefore seem to know many things, when they are for the most part ignorant and hard to get along with, since they are not wise, but only appear wise.*"

Socrates on writing[1]

Humankind has seen massive disruption since the advent of the steam engine, electricity, and the internet. These technologies profoundly changed society, but their impact on human cognition was indirect. Cognitive tools, too, have emerged in the last 500 years, like books, calculators, and search engines. Whenever a technology has become widespread, concerns have been raised about its impact on cognitive function and mental health. The telegraph was alleged to cause neurasthenia; TV and radio were linked with thought broadcasting and behavioural problems; the internet with addiction, and 'digital autism'.[2] It might seem far-fetched that even books were accused of being degrading according to ancient Greeks.[1] Are the concerns raised about the use of generative AI similar to those previous generations had, or is there a qualitative difference in the challenges faced by the human mind in the face of generative AI?

The generative AI revolution is driven by large language models (LLMs), which are based on the Transformer architecture.[3] These are next-word prediction algorithms, trained on a massive corpus of internet data and enhanced through techniques such as reinforcement learning from human feedback. LLMs interact with humans as chatbots, agents, copilots, APIs, and embedded assistants. They have become widely adopted at an unprecedented speed in translation, writing, programming, scientific research, education, finance, and healthcare.

Generative AI differs from the earlier cognitive tools. These are systems that help in dynamic two-way communications by mimicking human languages and emotions. They simulate

agency that impacts users in a way different from other non-interactive technologies. Humans tend to anthropomorphize them, attributing human-like understanding and emotions to text-based conversational agents.[4]

The previous technologies extended human cognition by storing and processing information. But the current generative AI technologies are increasingly participating in human reasoning, communication, and decision-making through natural language. This gives them characteristics of interactive cognitive agents. Humans may increasingly perceive them less as machines and more as active cognitive partners. We discuss this perceived active nature of AI and its impact on human cognition, language, and psychopathology. We argue that the recently proposed concept of cognitive surrender provides a useful framework for understanding these effects.

Cognitive offloading and cognitive surrender

With the advent of LLMs, the barriers to knowledge have lowered. People with limited access to educational resources can now acquire knowledge and skills more quickly. In addition to retrieving information, they help in creating ideas by recombining existing knowledge in new ways. Even in fields like chemistry and maths, LLMs are increasingly solving complex problems. It has become natural to first approach AI for a solution. Are human cognitive abilities enhanced or limited under continuous LLM use?

Calculators provide a useful analogy. When they were first introduced, teachers were against them, saying children would not learn the basics, and maths as a whole would be affected. However, they are now widely used, saving people from mundane math and helping them focus on more difficult problems. Similarly, books were initially accused of weakening the mind. But books played a major role in democratizing knowledge and even in the birth of the scientific age. Spreadsheets, GPS, and search engines have a similar story.

David Chalmers and Andy Clark put forward the concept of the extended mind, which argues that some physical objects aid cognition and should be considered part of the cognitive system.[5] This extends the mind beyond the confines of the brain and body. They state that objects such as sticky notes or calculators, which aid memory and maths, are properly understood as components of the mind. This delegation of certain cognitive tasks to external things is called cognitive offloading.

Such external aids leave evaluative reasoning and decisions largely under human control. They help with data collection and storage, but analysis and decision-making predominantly reside with humans. But with generative AI, humans increasingly defer reasoning and decision-making to AI-generated recommendations. This shifts the cognitive locus of control away from humans to the algorithm. The uncritical reliance on external reasoning that bypasses internal systems, thereby giving away the locus of control, has been termed cognitive surrender.[6]

Is cognitive surrender the default option in all humans who use it, or are there specific vulnerable groups? Given ongoing neurocognitive development and the formation of metacognitive and critical-thinking skills, we hypothesize that children may be more susceptible to this phenomenon. People with reduced curiosity or insufficient motivation for sustained cognitive effort may also be affected. On the other hand, individuals with greater intrinsic curiosity and those with high motivation for learning may be less prone.

Among active, reflective users, the impact of generative AI may be in augmenting their intellectual capabilities. Anecdotal reports from scientists suggest how AI is helping them form new conjectures and navigate the hypothesis space more efficiently. AI may also help improve creativity by finding connections in diverse fields. It has given tools to people who have never had mechanisms to express themselves.

Whether sustained interaction with AI causes neuroplastic changes in the brain remains unknown. Literacy has been shown to change human cortical networks for vision and language.[7] By analogy, sustained generative AI use may induce neuroplastic changes that enhance cognition and creativity in some people. Such changes are more likely to occur in children who adapt to newer technologies faster. Their innate curiosity and developmental plasticity may facilitate cognitive augmentation.

Thus, both cognitive surrender and cognitive augmentation could result from continued use of generative AI, especially in children. The actual outcome is likely to be mediated by user characteristics and task demands. The active, reflective users who decide to be the human in the loop will be treading the path towards augmentation, while the others may be surrendering their minds to AI.

Language and concepts

Generative AI is now widely used as a writing assistant. This has helped people draft good essays and emails, especially for those who write in their non-native language. AI helps to overcome writer's block and provides a good starting point for writing. Research has found that LLM assistance increased the novelty and usefulness of writing short stories, especially for less creative writers.[8]

The large-scale use of generative AI has made writing look similar and without individual style. LLM-written short stories are less diverse than human-written short stories.[9] It has been suggested that generative AI may enhance individual creativity at the cost of reduced collective creativity. Research has shown that though AI-generated stories are rated as more ingenious, they are more similar to each other than those written by humans.[8] The homogenization persists even after setting higher temperature settings or using prompts that ask the chatbot to take the simulated perspective of different identities.[9]

The use of a few LLMs to generate outputs for a multitude of different people results in homogeneity. Besides, LLMs give a seemingly final solution earlier in the work- these undermine variations in the later part of ideation. This has been described as creative fixation, where early solutions prevent exploration of potentially better alternatives.[10] This is analogous to optimisation getting trapped in a local minimum, blocking the search for better solutions. Neurophysiological studies have supported these findings. Preliminary evidence suggests that LLM-assisted writing is associated with reduced EEG connectivity, lower recall, and diminished subjective ownership of the work.[11]

Current evidence suggests that LLM outputs are biased towards Western writing styles and cultural norms, which cannot be explained away by grammar correction.[12] The homogenisation of language may contribute to epistemic homogeneity and later cultural and creative homogeneity.[8] We therefore speculate that increased reliance on generative AI may reduce linguistic diversity and thereby the range of conceptual frameworks for humans.

Psychopathology

Approximately 24% of US adults have sought relationship advice from AI instead of humans.[13] The users have a favourable opinion of the LLM chatbots as they are free and available when needed.[14] Anthropic reported that 2.9% of all conversations with Claude are affective in nature, seeking emotional support, advice, or companionship.[15] Many have argued that personalised LLM chatbots will be widely used as a relationship coach, and may end loneliness.[16]

Unlike traditional therapists, chatbots are available continuously and simultaneously engage millions of people. The accessibility may be of help for a large population with emotional distress. But this also brings psychological risk on an unprecedented scale. Many conversational AI systems are designed for prolonged use, which may reduce real-world interactions. Increasing dependence on AI relationships may adversely affect family and social bonds. Analysis of chatbot conversations has revealed themes revolving around sycophancy, romantic bonds, self-harm ideas, and delusional content. In some cases, the chatbot even encouraged violent behaviours.[17]

One important feature of AI-human interaction is sycophancy. Sycophantic AI agrees with the user regardless of its correctness. This arises because AI is trained via reinforcement learning from human feedback (RLHF)- AI that agrees with human beliefs is more likely to get positive feedback.[13] Susceptibility to sycophantic AI is not limited to the clinically vulnerable population. AI sycophancy is pervasive across current AI models, and most of the humans are susceptible to its effect.[18] Experimental evidence suggests that explicitly informing the users that the reply was AI-generated did not diminish its persuasive impact.[18]

A recent study found that AI chatbots reduce users' prosocial intentions, judgments, self-correction, repair of interpersonal conflicts, and responsibility taking.[18] They found that the chatbots affirmed users' actions even in cases involving illegality and deception. People who got advice from the AI were more likely to believe that they were correct in an interpersonal conflict, and less likely to apologize or repair the relationship. This suggests that AI sycophancy may worsen interpersonal relations by reinforcing the user's point of view rather than encouraging self-reflection and course correction.

The problem is amplified by the scale of LLM deployment. Chatbots are instantly available to millions of people, and their impact on human relationships can be enormous.

These built-in features of AI have clinical significance for vulnerable populations. People who have psychiatric illnesses or overvalued ideas rely on reality testing through their interaction with fellow human beings. A sycophantic AI chatbot may validate their overvalued ideas instead of challenging them. This reduces the chance of critical feedback to the human, resulting in worsening of symptoms. In such cases, LLMs are unlikely to be the primary cause of psychosis. They may act as an amplifier of the preexisting psychopathology. LLMs may even act as a coauthor in forming delusional narratives.[2]The reality distortion thus brought by the chatbot may affect treatment adherence in patients.

Though AI psychosis is widely used in media, it is not clinically validated.[13] Studies have shown that LLMs may become the subject of delusions.[2] This was seen in earlier technologies also. AI is described as a lover or God-like in clinical reports. This cannot be classified as psychosis caused by AI. The vulnerable population throughout history has incorporated the prevailing technology into their delusion. If this is the case, this represents historical continuity rather than a clinical novelty.

Both product- and user-level factors can worsen psychotic symptoms. Product-level factors include extended conversations, validating user beliefs, and simulating emotions and personas. User-level factors include sleep deprivation, social isolation, and vulnerable populations like psychotic patients, children, and adolescents.[13]

Our information about AI and its role in psychosis is minimal. Data about the definition, epidemiology, and whether LLMs are a causal factor are not clear. No prospective studies have been done. We are not sure if excessive LLM use is a coping mechanism to deal with stress or maladaptive, resulting in behavioural problems. This raises more questions, and only long-term research about whether and how LLM use causes behavioural and brain changes can identify vulnerable groups. What policy-level and engineering changes are needed to deal with this situation will become important in the future.

More than two millennia ago, Socrates questioned the then-current technology of writing, arguing that it weakened the human mind. However, over time, writing has transformed human society in ways that no other technology has. It has even rewired the human brain. Unlike writing, which preserves and transmits human knowledge, generative AI actively participates in knowledge generation. Whether the current concerns regarding AI will prove analogous to previous technological anxieties and whether LLMs ultimately reshape human cognition through processes of both augmentation and cognitive surrender can only be determined through longitudinal research. AI literacy, critical- thinking education, and human-in-the-loop practices may help build resilience against cognitive surrender.

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